Sushant Gupta

GitHub | GFG Profile | sushantgupta705@gmail.com| +91-6005067002 |LeetCode|

Education

Thapar Institute of Engineering & Technology July 2025 - June 2027
Masters of Technology in Computer Science and Engineering CGPA: 8.45/10.0

Global Academy Of Technology, Bengaluru Dec 2021 - June 2025
Bachelors of Technology in Information Science and Engineering CGPA: 9.13/10.0

St. Xavier's Convent school (CBSE), Kathua Apr 2020 – May 2021
12th Standard Percentage:87.6%

St. Xavier's Convent school (CBSE), Kathua Apr 2018 - Mar 2019
10th Standard Percentage:91.6%

Projects

HeartRiskX(Using Explainable Ai)

- Built HeartRiskX, an explainable AI system for heart disease risk prediction using ML models (Logistic Regression, Random Forest, XGBoost, LightGBM)
- Implemented a full **ML pipeline** including data preprocessing, feature encoding, threshold optimization, and performance evaluation
- Integrated **model explainability** using **SHAP and PDP/ICE** to provide transparent, interpretable predictions for clinical decision support.

Process Behavior Profiler (C++/Python)

- Developed a system profiling tool to monitor process-level CPU and memory usage over time and detect abnormal usage patterns
- Implemented time-series logging and anomaly detection for identifying potential performance bottlenecks
- Designed efficient data collection mechanisms for continuous system monitoring
- Applied system-level debugging and performance analysis techniques

Whistleblowing application

Technologies: MERN

- Unanimous Problem reporting: The user can file the complaint unanimously without his personal details being revealed.
- Implemented admin and user dashboards, where both can see and update the status
- The user can track the status with the generated one time code.

Thread Contention Visualizer (C++)

- Developed a simulation tool to visualize thread contention and synchronization issues in concurrent systems
- Implemented multithreading and synchronization mechanisms including mutexes and semaphores
- Analyzed deadlocks and resource contention scenarios to improve system performance
- Strengthened understanding of concurrency control and parallel execution

technical Skills

- **Programming Languages:** C++, Java, Python, JavaScript, Typescript
- **Technologies:** React, MongoDB, ExpressJs, flask, Nodejs, MySQL, Redux
- **Development tool:** Visual studio code ,Eclipse, Git and GitHub, Vite
- **Core Computer science:** Data Structures & Algorithms, Operating Systems, System security,System Programming Computer Architecture, OOP, Debugging, Process Management, Resource optimization, Computer Networks

Achievements

- **Certificates :** Acknowledgement certificates for java and web development, Oracle SQL(Foundation)
- Solved **300+** Questions on **DSA** (Data structures and Algorithms) on various platforms.